Examples of Poor IIR Filter Design

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Key Concepts

1) **Always** verify that your design satisfies the specifications.

2) Very tight specifications can lead to meaningless filter designs.

    a) Numerical computation errors can result in unacceptable designs.

    b) The design procedure is numerically sensitive when multiple poles or zeros are located in a small region.

c) Stringent specifications on passband ripple, transition-band width, and stop band attenuation may not be able to be satisfied with practical filters.

3) Signs of a possible poor design:

a) Large filter orders ($N >> 10$).

b) Large numerator or denominator coefficients (values $>>$ 50).

c) Any design with poles outside the unit circle is unacceptable.

d) Frequency response specifications are not satisfied.

4) Rescuing a poor design

a) Change prototype filter to one that is more easily able to meet tight specifications with a low order. For example, use Chebyshev I or II instead of Butterworth, or use elliptic instead of Chebyshev I or II.

b) Increase the width of the transition band.

c) Decrease the amount of stop band attenuation.

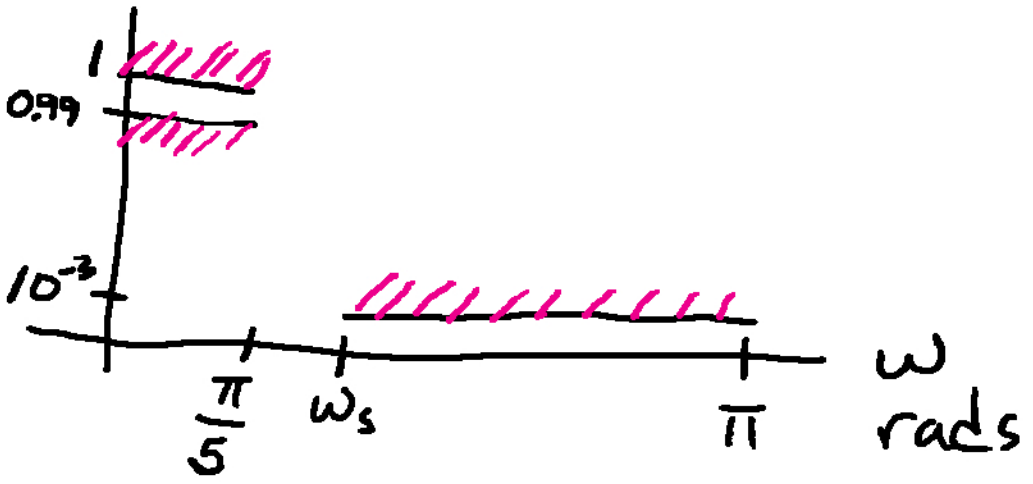
d) Increase the amount of passband ripple.

Video Screenshots

Always verify that your design satisfies the specifications!

Very tight specs can result in meaningless designs

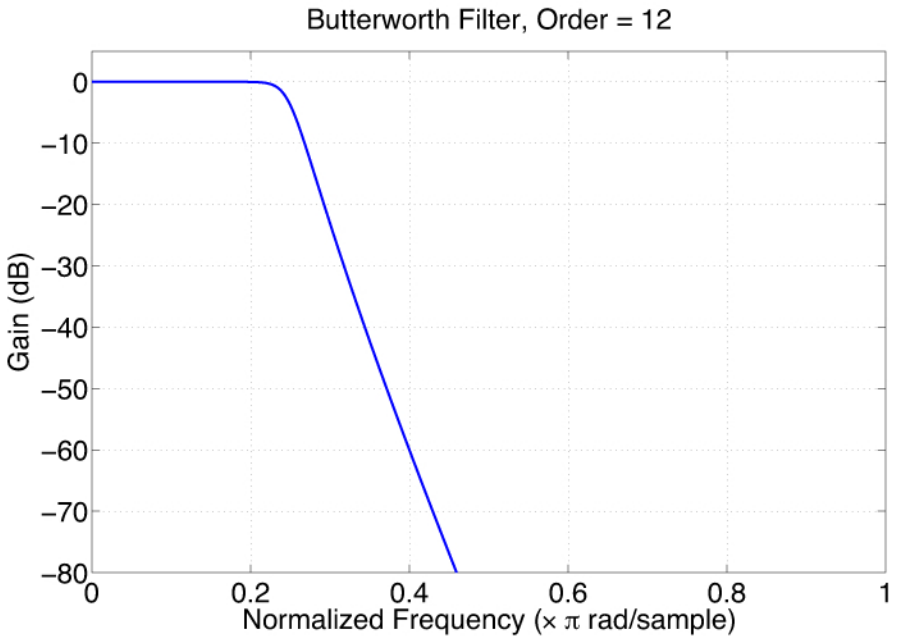
Example: Low pass Butterworth

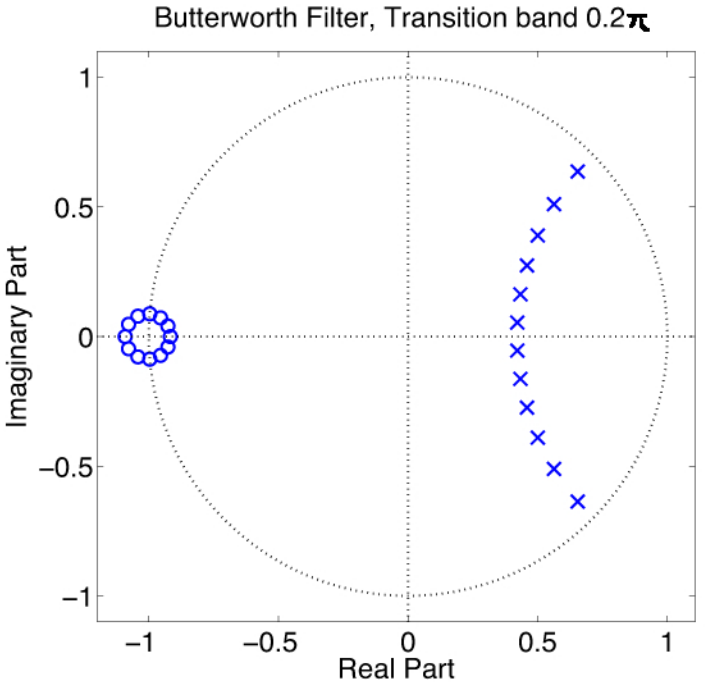


Case 1: $\omega_s = 0.4\pi$ rads

Case 2: $\omega_s = 0.24\pi$ rads

Case 1: Transition band 0.2π rads





Coefficients, transition band 0.2pi

B = 1.0e-03 *

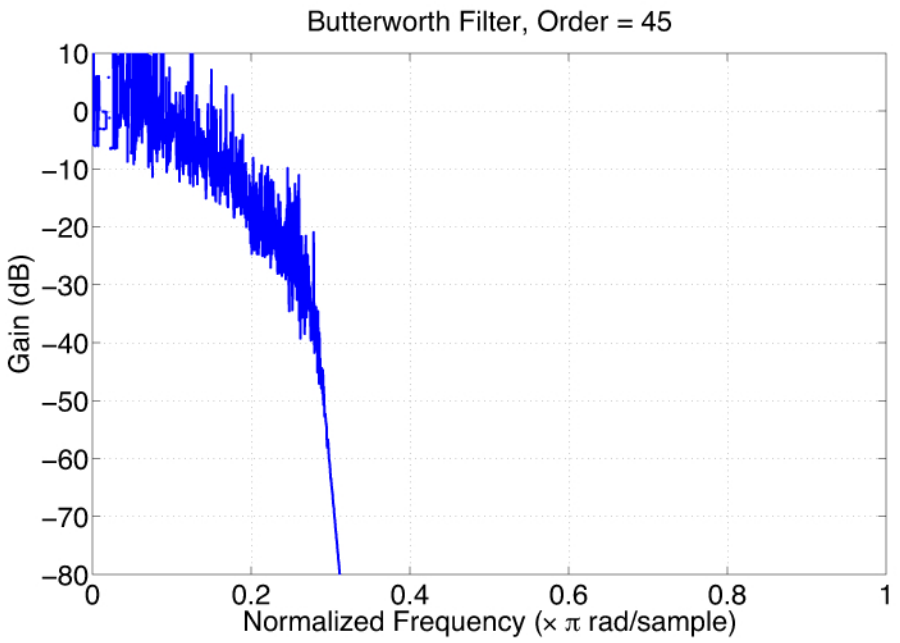
0.0010 0.0120 0.0660 0.2200 0.4950 0.7921 0.9241 0.7921 0.4950 0.2200 0.0660 0.0120 0.0010

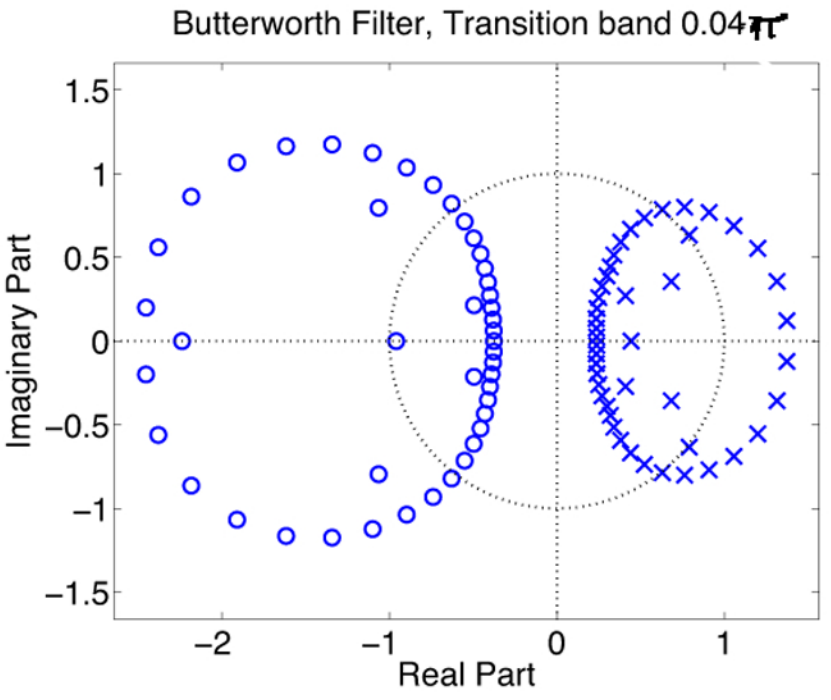
A =

1.0000 -6.0629 17.7293 -32.6911 42.0486 -39.5581 27.8126 -14.6843 5.7654 -1.6386 0.3195 -0.0383 0.0021

Case 2: Transition Band 0.04π rads

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Coefficients, transition band 0.04pi

B = 1.0e-07 *

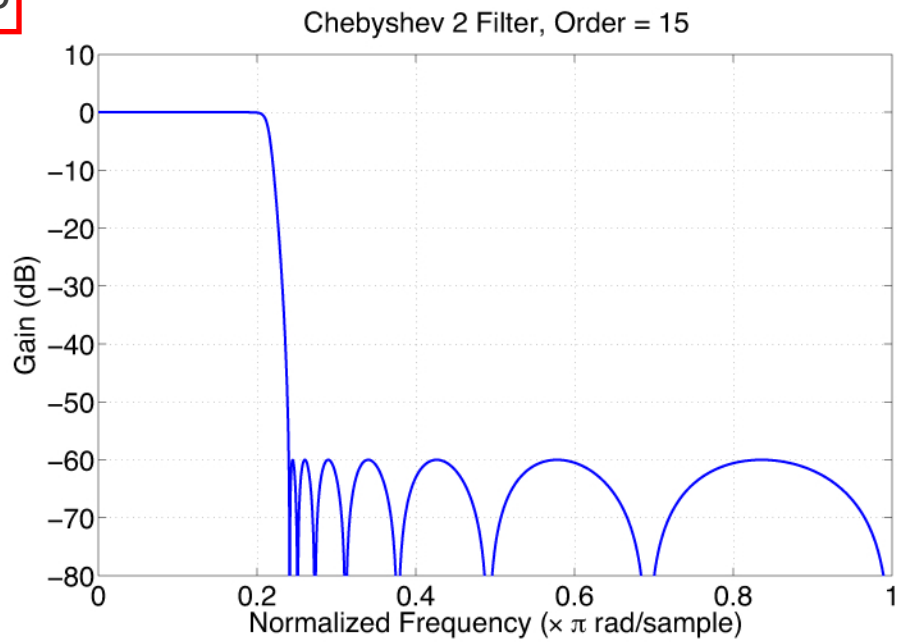
-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0001	-0.0003
-0.0010	-0.0025	-0.0057	-0.0117	-0.0219	-0.0374	-0.0581	-0.0826	-0.1074	-0.1279	-0.1395	-0.1395
-0.1279	-0.1074	-0.0826	-0.0581	-0.0374	-0.0219	-0.0117	-0.0057	-0.0025	-0.0010	-0.0003	-0.0001
-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000		

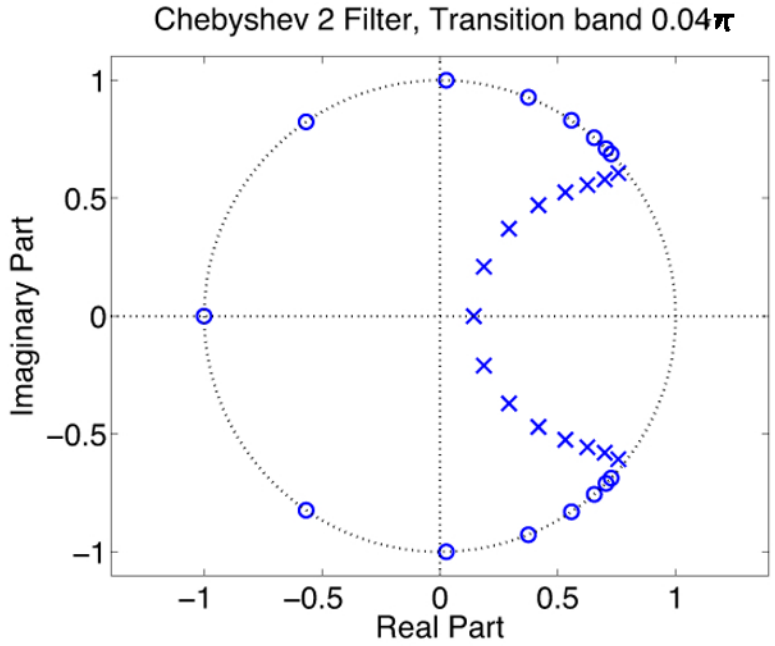
A = 1.0e+08 *

0.0000	-0.0000	0.0000	-0.0000	0.0002	-0.0009	0.0036	-0.0121	0.0352	-0.0889	0.1981	-0.3923
0.6953	-1.1100	1.6037	-2.1052	2.5192	-2.7555	2.7609	-2.5383	2.1441	-1.6655	1.1904	-0.7831
0.4740	-0.2640	0.1351	-0.0635	0.0274	-0.0108	0.0039	-0.0013	0.0004	-0.0001	0.0000	-0.0000
0.0000	-0.0000	0.0000	-0.0000	0.0000	-0.0000	0.0000	-0.0000	0.0000	-0.0000		

Chebyshev 2 design: Transition band 0.04π rads 5

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Chebyshev 2 Coefficients

B =

0.0029 -0.0114 0.0285 -0.0461 0.0577 -0.0528 0.0377 -0.0122 -0.0122 0.0377 -0.0528 0.0577
-0.0461 0.0285 -0.0114 0.0029

A =

1.0000 -7.1645 25.3133 -57.6380 93.8983 -115.3399 109.9630 -82.6252 49.2266 -23.2124 8.5801 -2.438
0.5151 -0.0763 0.0071 -0.0003

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